Mannava Sainadh Charan

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Summary

Backend-focused software engineer building AI-powered products and scalable backend systems. Experienced with Python, FastAPI, Java, Spring Boot, REST APIs, MongoDB, SQL, and JWT-secured services. Currently optimizing AI-agent workflows and backend services for an AI fitness application and an animated cricket game for mobile and tablet platforms.

Education

Koneru Lakshmaiah Education Foundation (KL University) Vaddeswaram, Andhra Pradesh
B.Tech in Electronics and Computer Science – CGPA: 8.92 *2022 – 2026*

Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Distributed Systems, Software Engineering, Systems Design, Machine Learning

Technical Skills

Languages: Java, Python, JavaScript, SQL
AI/ML: LangChain, LangGraph, RAG, Agentic Workflow Design, Vector Databases (ChromaDB), CNNs
Backend & APIs: FastAPI, Spring Boot, Django, REST APIs, MVC, CRUD, API Design
Databases: MongoDB, MySQL, PostgreSQL, SQLAlchemy, Redis
Security: JWT Authentication, Authorization, Input Validation, Error Handling
Tools: Git, GitHub, Maven, Docker, Postman, VS Code

Experience

Backend Developer – Ranktel Studios Bengaluru, Karnataka
AI Fitness & Mobile Gaming *September 2026 – Present*

- Building backend systems for the **Forge Club AI Fitness App** and an **animated cricket game** for mobile and tablet platforms.
- Optimizing **AI-agent workflows** and backend services supporting AI-powered fitness features, user profiles, workout tracking, and application intelligence.
- Developing **RESTful APIs using Python and FastAPI**, including authentication, validation, error handling, and business logic.
- Designing and managing application data using **MongoDB and SQL databases** based on feature and scalability requirements.
- Developing backend services for **player profiles, game data, and gameplay-related features** for the animated cricket game.

Web Developer Intern – iNeuBytes Vijayawada, Andhra Pradesh
Backend & Payments, RR Durbar Restaurant Ordering Platform

- Built and maintained backend REST API endpoints powering cart, checkout, and order management for a live multi-outlet restaurant ordering system.
- Implemented and tested online payment integration end-to-end, including transaction requests, payment status updates, and order confirmation logic.
- Added input validation and error handling on payment and order endpoints; collaborated with the team to debug production issues and ship fixes.

Projects

Real-Time Facial Emotion Detection System | *Python, OpenCV, Neural Networks*

- Built a real-time emotion-detection pipeline that processed live video and classified emotions using computer vision and neural networks.
- Improved classification accuracy through iterative testing and debugging, turning an initial prototype into a reliably working system.

Self-Correcting RAG Pipeline | *Python, LangGraph, LangChain, Vector Retrieval*OneinBox AI Internship Hackathon 2026

- Built a backend orchestration system that identified weak retrievals and corrected them through query rewriting in a closed feedback loop.
- Designed the six-layer architecture and implemented a five-stage pipeline covering retrieval, grading, routing, generation, and verification.

Manpower Acquisition & Employee Recruitment System | *Java, Spring Boot, REST APIs, MySQL*

- Developed a secure recruitment platform supporting job posting, candidate tracking, and recruiter management using Spring Boot REST APIs and MySQL.
- Implemented JWT-based authentication and authorization to protect sensitive recruiter and candidate data.

Certifications

Red Hat Enterprise Application Developer | Microsoft Azure AZ-900 | IBM Python for Everyone

Achievements

Finalist, GeeksforGeeks Hackathon 2025, SRM AP | Secured AIR 67 in B.Arch entrance exam, 2022